

ME-255 A:FLUID MECHANICS I (M)

For

**B.E II YEAR IV SEMESTER
(MECHANICAL ENGINEERING)**



By

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(UNIT -3)

FLUID KINEMATICS

LECTURE 12

Fluid Kinematics

"Fluid Kinematics deals with study of velocity and acceleration of fluid particles and their distribution in space without consideration of any forces involved."

velocity and acceleration at a point in a fluid can be denoted as

$$\text{Velocity vector} = \vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$$

where u , v and w are the components of velocity in x , y and z directions, respectively.

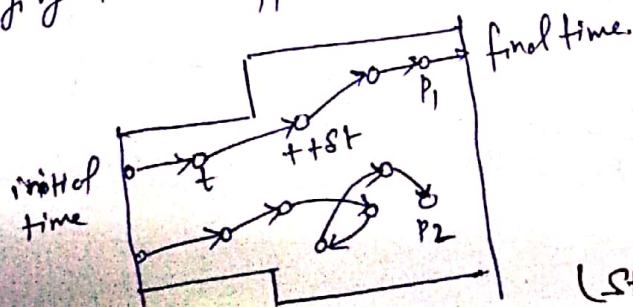
$$\text{Similarly } \vec{a} = a_x\vec{i} + a_y\vec{j} + a_z\vec{k}$$

where a_x , a_y and a_z are the rectangular components of acceleration in x , y and z directions, respectively.

Lagrangian and Eulerian approach

Fluid motion can be studied using two approaches.

In Lagrangian approach, fluid particle is identified and followed during course of motion with time. The particle may change its shape, size and state as it moves. Difficulty in tracing a single fluid particle rules out the possibility of applying this approach.



(Study of each particle with time)
(Lagrangian approach)

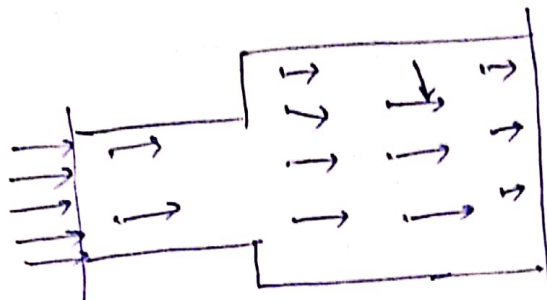
In Eulerian approach fluid motion is studied at fixed points in flow field. motion of fluid is specified by velocity components as a function of space and time.

$$u = u(x, y, z, t)$$

$$v = v(x, y, z, t)$$

$$w = w(x, y, z, t)$$

This approach is applied in study of fluid mechanics



Study at all points
a time
(Eulerian approach)

Types of fluid flow

Steady and Unsteady flow

A flow is considered to be steady if dependent fluid variables (eg. velocity, density, viscosity, pressure and temperature) at a point in flow do not change with time.

i.e.

$$\frac{\partial (\text{dependent fluid variable})}{\partial t} = 0$$

$$\Rightarrow \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{\partial w}{\partial t} = 0 = \frac{\partial p}{\partial t} = \frac{\partial \rho}{\partial t}$$

A flow is considered as unsteady if the dependent fluid variables at any point in flow changes with the passage of time.

$$\frac{\partial (\text{dependent fluid variable})}{\partial t} \neq 0$$

- Steadiness of flow implies that flow pattern does not change with time.
- Unsteadiness refers to changing flow pattern with passage of time at same points in space.

Uniform and Non uniform flow

A flow is said to be uniform at any instant of time if the velocity components are identical throughout the flow field. i.e. velocity components are same at different locations in flow.

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial v}{\partial z} = 0$$

The flow is considered non uniform if velocity components at given instant of time are different at different locations.

$$\frac{\partial (\vec{V})}{\partial s} \neq 0$$

- "Conventionally uniformity of flow is defined in terms of velocity components rather than all dependent fluid variables"
- Steadiness refers to constancy with respect to "time" and Uniformity refers to constancy with respect to "space"
- Flow can be steady or unsteady independently of being its uniform or non-uniform.

LECTURE 13

- All four combinations are possible.

(a) Steady and uniform: Flow in straight long pipe of constant diameter with constant discharge.

(b) Steady and non-uniform: flow in a conical pipe having constant discharge.

(c) Unsteady and uniform: flow in a straight long pipe of constant diameter with varying discharge.

(d) Unsteady and nonuniform: flow in a conical pipe having varying discharge.

One, two and three dimensional flow

"A fluid flow is said to be one, two and three dimensional if Velocity component is the function of one, two and three coordinates, respectively."

Symbolically

Steady

Unsteady

One-dimensional flow: $\bar{V} = f(x)$

$$\bar{V} = f(x, t)$$

Two dimensional flow: $\bar{V} = f(x, y)$

$$\bar{V} = f(x, y, t)$$

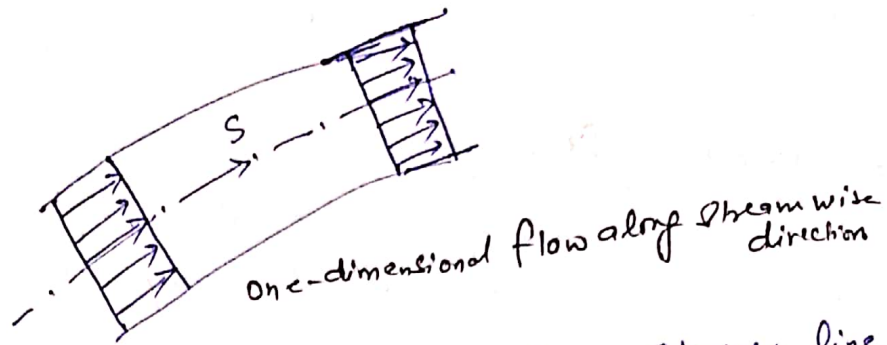
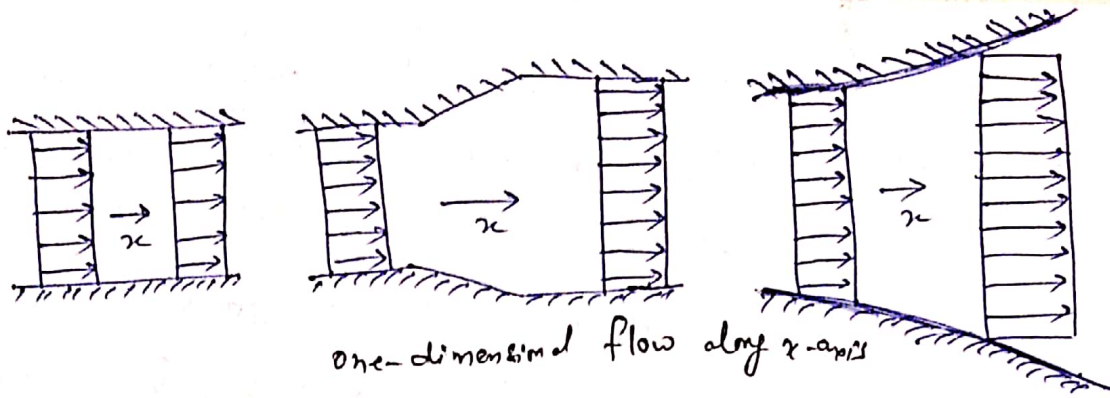
Three dimensional flow: $\bar{V} = f(x, y, z)$

$$\bar{V} = f(x, y, z, t)$$

One-dimensional flow

- In this flow the dependent variables are only function of one space coordinate eg. x .

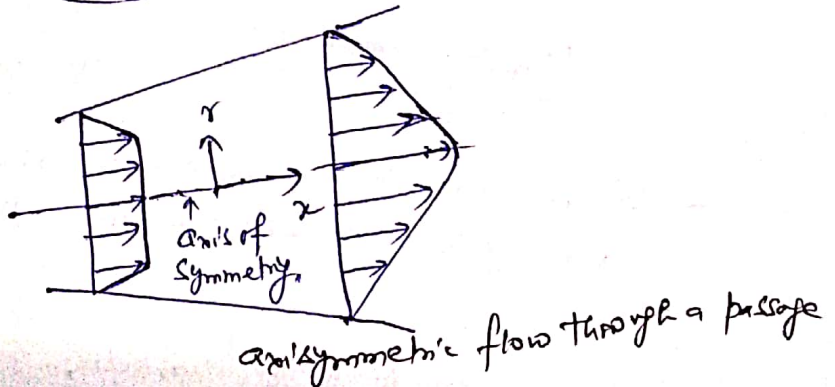
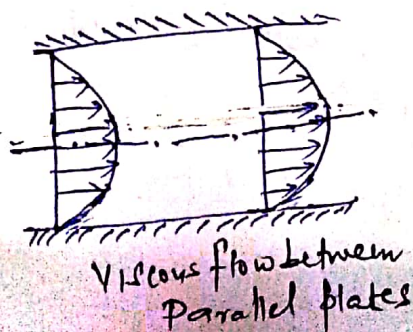
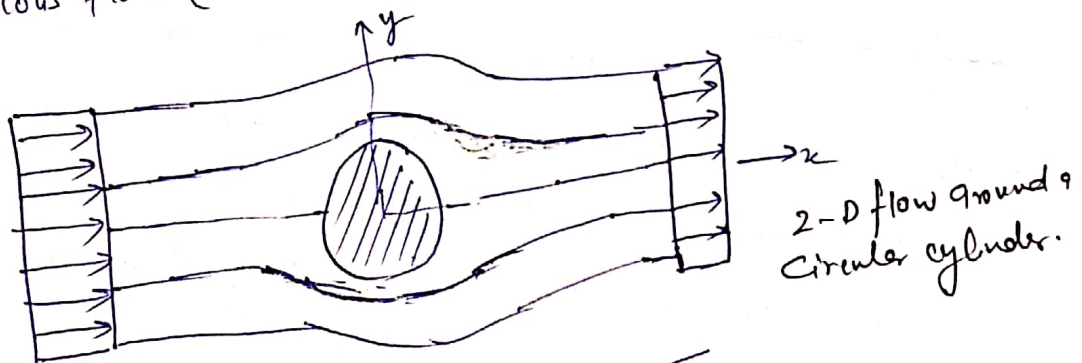
- eg. flow through pipes, channels and variable area ducts if velocity distribution is considered constant at each cross section.



- The axis of flow does not have to be a straight line for a flow to be one dimensional.

Two-dimensional flow

- In this flow dependent variables are function of two space coordinates.
- This means flow is identical in parallel planes.
- examples are flow around a circular cylinder of infinite length, viscous flow (real fluid) through pipes, between parallel plates etc.



- A flow is called axisymmetric if velocity profile is symmetrical about axis of symmetry. i.e. velocity profile is same at different diametral planes drawn through passage, i.e. No variation of velocity profile with respect to θ
- An axisymmetric flow is essentially 2-D flow, $\bar{V} = f(r, x)$

Three-dimensional flow

- In three dimensional flows flow properties vary in all three directions such as flow in a river, flow within fluid machines etc.

Laminar flow and turbulent flows

A Laminar flow is a flow in which fluid particles move in thin layers (laminae) with one layer sliding over other. The particles move in definite and observable paths. There is no exchange of fluid particles from one layer to another. This flow is characterized by small velocity,

A turbulent flow is a flow in which all fluid particles are distributed and there is continuous transfer of momentum to adjacent layers. In this flow, the velocity varies from point to point in magnitude and direction as well as from time to time.

Two flows are characterized by Reynold's number

$$\text{Reynolds number (Re)} = \frac{\text{Inertia force}}{\text{Viscous force}}$$

- Viscous force is frictional force due to viscosity of fluid and try to make fluid motion in parallel layers.
- Inertia force is due to mass and velocity of fluid particles and try to diffuse ^{the} particles.

- At low velocities, the inertia forces are less in comparison to viscous forces and the flow is laminar.
- But when the velocity is high, fluid layers break due to predominance of inertia forces and flow is turbulent.

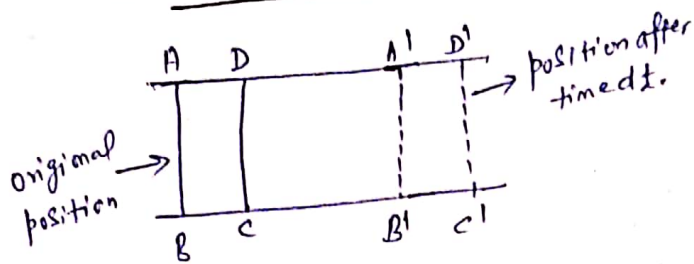
for pipe flow

$Re < 2100$ (laminar flow)

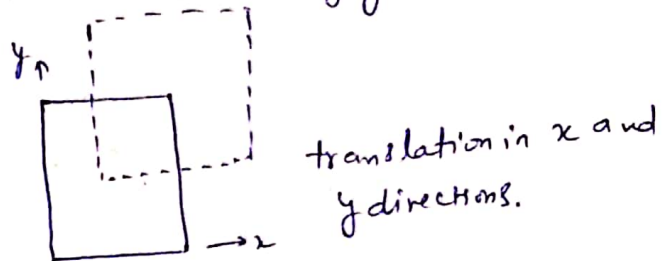
$Re > 3000$ (turbulent flow)

Different types of displacement of fluid particles

(a) Linear translation or pure translation or irrotational translation

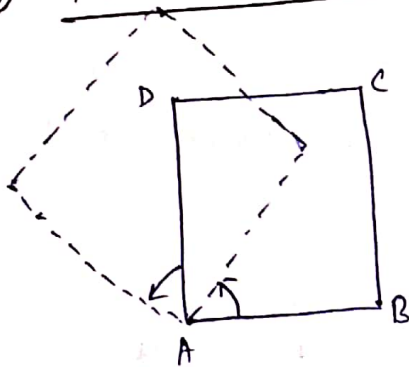


- The element moves as a body without changing its dimensions.



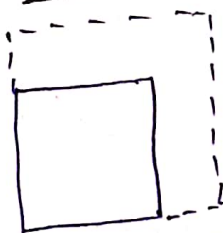
translation in x and y directions.

(b) Pure rotation or rotational translation

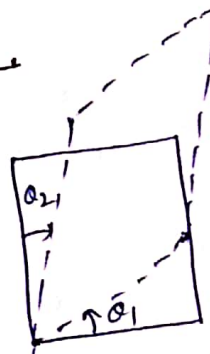


Rotation of AB and AD axes in same direction.

(c) Pure distortion or deformation



Volume or linear deformation

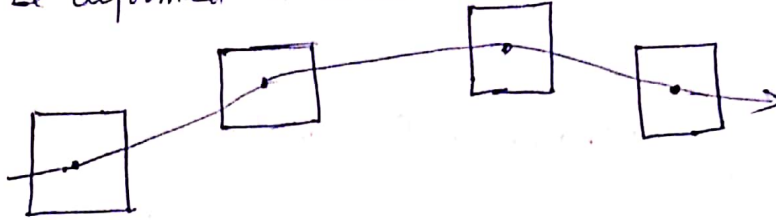


Angular or shear deformation

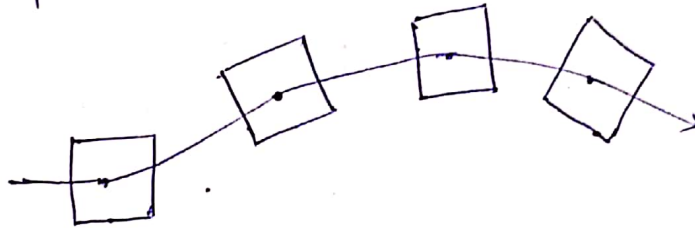
LECTURE 14

Rotational flow and Irrotational flow

An irrotational flow is one in which fluid element has no net rotation about its own axis which is perpendicular to plane of motion while moving in the direction of flow. The fluid element may be deformed or distorted.



If net rotation of fluid element about its own axis is not zero and they rotate while moving in the direction of flow, then the flow is termed as rotational flow.

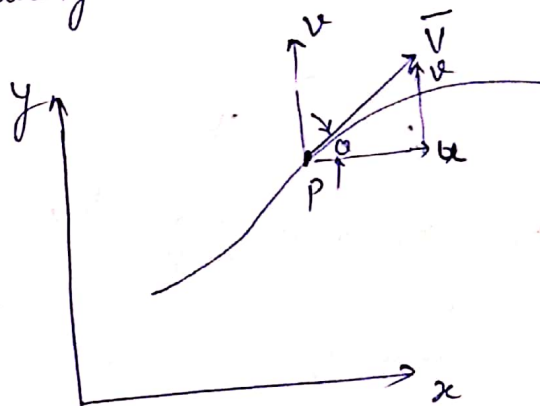


Streamlines, Pathlines and Streaklines

"A Streamline in a fluid flow is a line tangent to which at any point is the direction of velocity at that point at that instant."

By definition

velocity vector $\vec{V}$ at point P must be tangential to streamline at that point.



$$\tan \theta = \frac{dy}{dx} = \frac{v \cdot dt}{u \cdot dt}$$

Stream-line in x-y plane

$$\boxed{\frac{dx}{u} = \frac{dy}{v}}$$

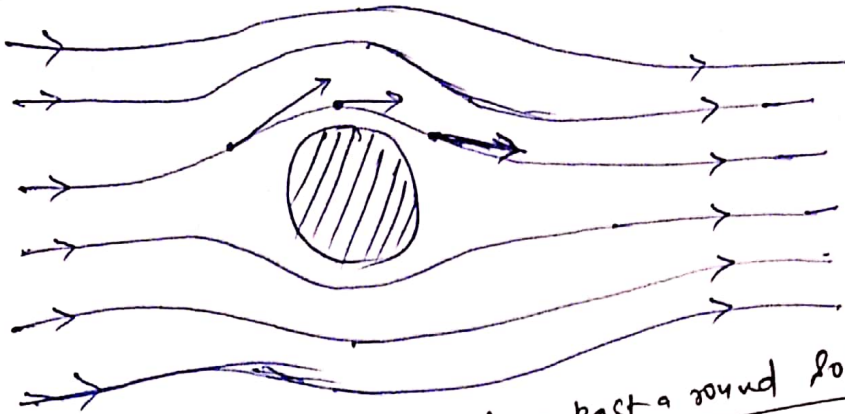
$dt = \text{small time interval}$
General equation of Streamline in three-dimensional flow

$$\boxed{\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}}$$

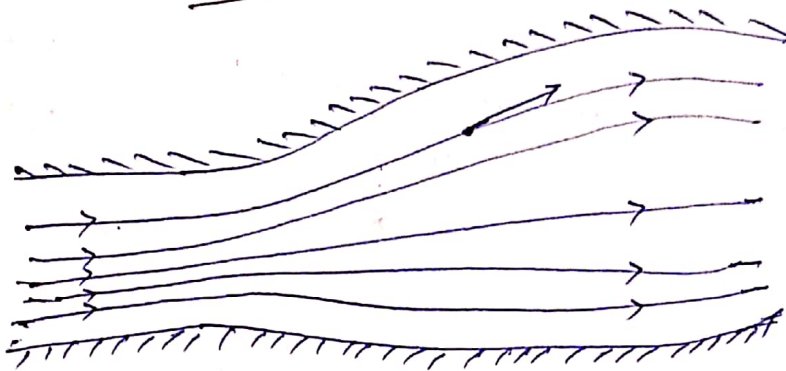
$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Valid for

- Steady or unsteady
- Uniform or nonuniform
- viscous or inviscid
- compressible or incompressible



Streamlines of flow past a round solid object



Streamlines of flow within a close conduit

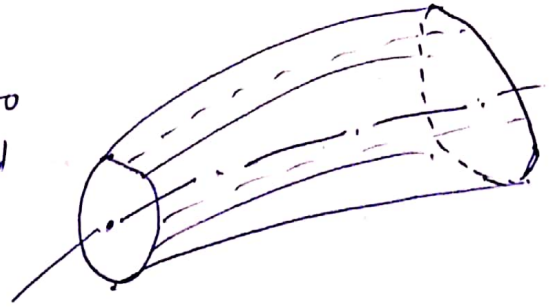
Some facts

- Streamlines are equivalent to instantaneous snap-shot indicating the direction of velocity in the entire flow field.
- Since the velocity is tangent at any point on the streamline, hence the component of velocity normal to streamline is zero, therefore there can be no flow across the streamline.
- No two streamlines can cross (intersect) each other because at the point of intersection fluid particle will have two velocity and that is physically impossible.

Stream tube

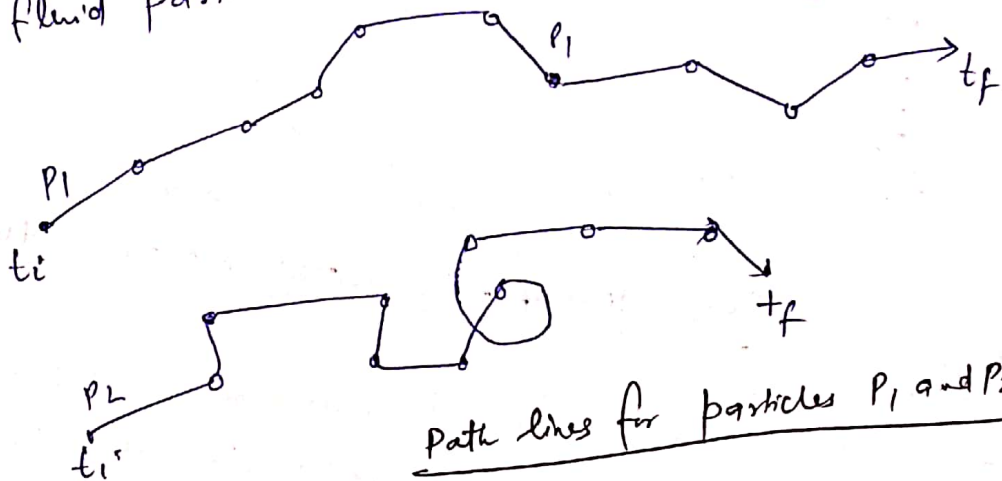
"The stream tube may be defined as a tubular space formed by collection of streamlines passing through the perimeter of a closed curve"

- Since the velocity component normal to streamlines is zero hence no fluid can enter or leave the stream tube except at ends.



Path line

"A path line in a fluid flow is the path traced by a single fluid particle in motion over a period of time." $(t_i \text{ to } t_f)$



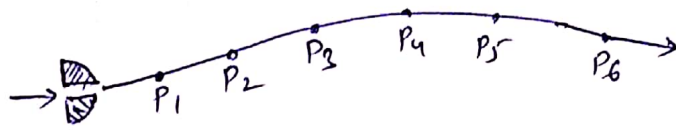
path lines for particles P1 and P2

- Tangent at a point on a path line must be the direction of velocity at that point at the time when the particle passes that point. A path line can therefore intersect itself at different times. In general path lines constitute a pattern of intersecting zig-zag lines.
- In steady flow velocity at each point remains unaltered with time. Hence streamlines pattern does not change with time. Individual fluid particles must follow the streamlines, hence for steady flow, streamlines and path lines coincide.

Streak line

"Streak line is the instantaneous picture of the positions of all the fluid particles that have passed through a fixed point in the flow field."

- example is lines formed by smoke particles ejected in the flow from a fixed nozzle is a streak line.



Streak line for particles
 $P_1, P_2, \dots$
(at $t = t_1$)

- In steady flow stream lines, path lines and streak lines coincide.

Tutorial problem (2d wed)

In a flow the velocity vector is given by.

$$\vec{V} = 3x\vec{i} + 4y\vec{j} - 7z\vec{k}$$

Determine the equation of stream line passing through a point (1, 4, 5).

Solution The equation of stream line is

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

$$\frac{dx}{3x} = \frac{dy}{4y} = \frac{dz}{-7z}$$

Considering equation involving x and y

$$\frac{1}{3} \int \frac{dx}{x} = \frac{1}{4} \int \frac{dy}{y} \Rightarrow \frac{1}{3} \ln x = \frac{1}{4} \ln y + \ln C_1'$$

C_1' = a Constant of integration

$$\ln(x^{1/3}) = \ln(y^{1/4}) + \ln C_1'$$

$$\ln x^{1/3} = \ln(C_1' y^{1/4})$$

$$= \frac{1}{4} = \frac{C_1' y^{1/4}}{x^{1/3}}$$

$$\Rightarrow y = C_1 x^{4/3}$$

C_1 = another Constant

Considering equation involving x and z

$$\frac{dx}{3x} = \frac{dz}{-7z} \Rightarrow \frac{1}{3} \ln x = -\frac{1}{7} \ln z + \ln C_2'$$

$$\ln(x^{1/3}) = \ln(z^{-1/7}) + \ln C_2'$$

$$\ln(x^{1/3}) = \frac{C_2' z^{-1/7}}{C_2'}$$

$$z = C_2 x^{-7/3}$$

C_2 = another Constant

point (1, 4, 5)

$$y = C_1 x^{4/3} \Rightarrow 4 = C_1 (1)^{4/3} \Rightarrow C_1 = 4$$

$$z = C_2 x^{-7/3} \Rightarrow 5 = C_2 (1)^{-7/3} \Rightarrow C_2 = 5$$

Hence stream line passing through (1, 4, 5) is

$$\boxed{\begin{matrix} y = 4 x^{4/3} \\ z = 5 x^{-7/3} \end{matrix}}$$

LECTURE 15

Velocity and Acceleration

Let $\vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$ is the velocity vector at any point in the fluid flow.

Velocity components are the functions of space and time.

$$u = u(x, y, z, t)$$

$$v = v(x, y, z, t)$$

$$w = w(x, y, z, t)$$

Using differential calculus, infinitesimal change in u is given by:

$$\delta u = \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z + \frac{\partial u}{\partial t} \delta t$$

Let a_x , a_y and a_z be total acceleration ^{components} in x , y and z directions, respectively.

$$a_x = \frac{\delta u}{\delta t} = \frac{\partial u}{\partial x} \frac{\delta x}{\delta t} + \frac{\partial u}{\partial y} \frac{\delta y}{\delta t} + \frac{\partial u}{\partial z} \frac{\delta z}{\delta t} + \frac{\partial u}{\partial t}$$

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}$$

$$a_y = \frac{\delta v}{\delta t} = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}$$

$$a_z = \frac{\delta w}{\delta t} = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t}$$

Similarly

Total acceleration

Convective acceleration

i.e.

instantaneous space rate of change of velocity

local acceleration,

i.e.

local time rate change of velocity.

Total

acceleration vector $\vec{a}$ is given as

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

Tutorial problem (Solved)

The velocity vector in an incompressible flow is given by

$$\vec{V} = (6xt + yz^2)\mathbf{i} + (3t + xy^2)\mathbf{j} + (xy - 2xyz - 6tz)\mathbf{k}$$

(i) Verify whether the continuity eqn is satisfied?

(ii) Determine the acceleration vector at point A (1,1,1) at $t=1.0$.

Solution

$$u = 6xt + yz^2, \quad v = 3t + xy^2, \quad w = xy - 2xyz - 6tz$$

$$\frac{\partial u}{\partial x} = 6t, \quad \frac{\partial v}{\partial y} = 2xy, \quad \frac{\partial w}{\partial z} = -2xy - 6t$$

(i) from Continuity eqn, $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 6t + 2xy - 2xy - 6t = 0$
Hence continuity eqn is satisfied.

$$\begin{aligned} (ii) \quad a_x &= \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \\ &= 6x + (6xt + yz^2)(6t) + (3t + xy^2)(2xy) + (xy - 2xyz - 6tz)(yz) \\ a_y &= \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \\ &= 3 + (6xt + yz^2)(y^2) + (3t + xy^2)(2xy) + (xy - 2xyz - 6tz)(0) \\ a_z &= \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \\ &= -6z + (6xt + yz^2)(y - yz) + (3t + xy^2)(x - 2yz) \\ &\quad + (xy - 2xyz - 6tz)(-xy - 6t) \end{aligned}$$

at point (1,1,1) and $t=1.0$

$$a_x = 6 + (6+1)(6) + (3+1)(1) + (1-2-6)^2 = 6 + 42 + 4 - 14 = 38 \text{ units}$$

$$a_y = 3 + (6+1)(1) + (3+1)(2) + (1-2-6)(0) = 18 \text{ units}$$

$$a_z = -6 + (6+1)(1-2) + (3+1)(1-2) + (1-2-6)(-2-6) = 39 \text{ units}$$

$$\vec{a} = \underline{\underline{38\mathbf{i} + 18\mathbf{j} + 39\mathbf{k}}}$$

Tutorial Problem (Solved)

A fluid flow field is given by

$$\vec{V} = x^2y \mathbf{i} + y^2z \mathbf{j} - (2xyz + yz^2) \mathbf{k}$$

Prove that this is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point $(2, 1, 3)$.

Solution:

$$u = x^2y, \quad v = y^2z, \quad w = -(2xyz + yz^2)$$

$$\frac{\partial u}{\partial x} = 2xy, \quad \frac{\partial v}{\partial y} = yz, \quad \frac{\partial w}{\partial z} = -2xy - yz$$

from Continuity eq. $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 2xy + yz - 2xy - yz = 0$
i.e. possible case.

$$\vec{V} = (2)^2 \cdot 1 \mathbf{i} + 1^2 \cdot 3 \mathbf{j} - (2 \cdot 2 \cdot 1 \cdot 3 + 1 \cdot 3^2) \mathbf{k}$$

$$= 4\mathbf{i} + 3\mathbf{j} - 21\mathbf{k} \quad \text{ms}$$

velocity magnitude $V = \sqrt{u^2 + v^2 + w^2} = \sqrt{4^2 + 3^2 + (-21)^2} = \sqrt{466} = \underline{\underline{21.587 \text{ ms}}}$

acceleration $a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = x^2y(2xy) + y^2z(2^2) - (2xyz + yz^2) \cdot 0$
 $= 2x^3y^2 + x^2y^2z = 2 \cdot 2^3 \cdot 1^2 + 2^2 \cdot 1^2 \cdot 3$
 $= 16 + 12 = \underline{\underline{28}}$

$a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = (x^2y) \cdot 0 + y^2z(yz) - (2xyz + yz^2) \cdot y$
 $= y^3z^2 - 2xy^3z - y^3z^2$
 $= 2 \cdot 1^3 \cdot 3^2 - 2 \cdot 2 \cdot 1^3 \cdot 3 - 1^3 \cdot 3^2$
 $= 18 - 12 - 9 = \underline{\underline{-3}}$

$a_z = u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = (x^2y)(-yz) + y^2z(-2xz - yz^2)$
 $+ (-2xyz - yz^2)(-2xy - yz)$
 $= -2x^3yz - 2xy^3z - y^3z^3$
 $+ 4x^2yz^2 + 4xy^2z^2 + 2xy^2z^3$
 $+ 2y^2z^3$

$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$
 $a_z = 1^2 \cdot 3 [2x^2 + 3 + 4 \cdot 2 \cdot 3]$
 $= 3 [8 + 9 + 24]$
 $= 61 \cdot 3$
 $\underline{\underline{a_z = 123}}$

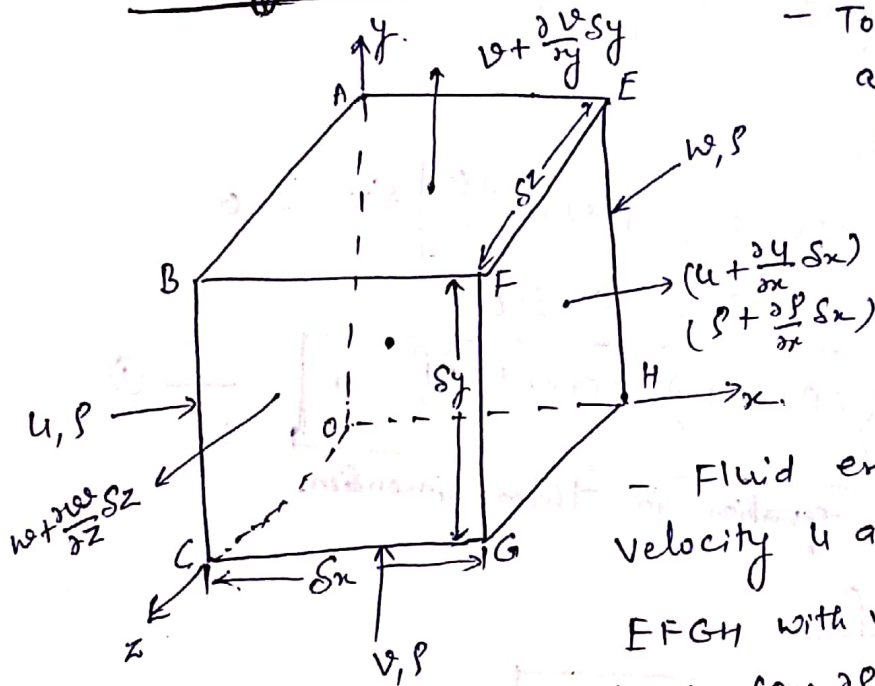
$a = \sqrt{28^2 + (-3)^2 + 123^2} = \sqrt{12618} \text{ ms}^2$
 $= y^2z [2x^2 + z^2 + 4xz]$

LECTURE 16

Continuity equation

Continuity equation is derived on the principle of Conservation of mass, according to which in a non nuclear process matter can neither be created nor destroyed. This means mass of fluid per unit time entering at one end of control volume is equal to mass of fluid leaving at other end of control volume plus mass of fluid accumulated in control volume.

Continuity equation in three dimension for Cartesian coordinates



- To derive the Continuity equation at a point in a fluid, a control volume in form of rectangular parallelepiped is considered

- Fluid entering face ABCD with velocity u and density ρ leave face EFGH with velocity $(u + \frac{\partial u}{\partial x} \Delta x)$ and density $(\rho + \frac{\partial \rho}{\partial x} \Delta x)$

$$\text{mass rate entering } ABCD = \rho u \Delta y \Delta z$$

$$\begin{aligned} \text{mass rate leaving } EFGH &= (\rho + \frac{\partial \rho}{\partial x} \Delta x) (u + \frac{\partial u}{\partial x} \Delta x) \Delta y \Delta z \\ &= \rho u \Delta y \Delta z + \rho \frac{\partial u}{\partial x} \Delta x \Delta y \Delta z + u \frac{\partial \rho}{\partial x} \Delta x \Delta y \Delta z \\ &\quad + \text{smaller terms} \end{aligned}$$

$$\begin{aligned} \text{net mass leaving } x\text{-direction} &= \\ &= (\rho \frac{\partial u}{\partial x} + u \frac{\partial \rho}{\partial x}) \Delta x \Delta y \Delta z \\ &= \frac{\partial}{\partial x} (\rho u) \delta V \end{aligned}$$

$$\delta V = \text{Volume of Control Volume}$$

net mass leaving y-direction
 $= \frac{\partial}{\partial y} (\rho v) \delta V$

net mass leaving z-direction
 $= \frac{\partial}{\partial z} (\rho w) \delta V$

Rate of accumulation of mass within control volume is given by

$$\frac{\partial}{\partial t} (\rho \delta x \delta y \delta z) = \frac{\partial \rho}{\partial t} \delta V$$

by mass balance

$$\left[\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) + \frac{\partial \rho}{\partial t} \right] \delta V = 0$$

Since $\delta V \neq 0$

$$\boxed{\frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) + \frac{\partial \rho}{\partial t} = 0} \quad \text{--- ①}$$

This is continuity equation in three dimension.

In vector form.

$$\boxed{\nabla \cdot (\rho \vec{U}) + \frac{\partial \rho}{\partial t} = 0}$$

$$\nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\vec{U} = u \vec{i} + v \vec{j} + w \vec{k}$$

Equation ① is Continuity equation

- ① In three dimensions for general flow
- ② For compressible or incompressible fluids, because ρ variation is taken into account.

3

(c) For steady or unsteady flow because variation with respect to time is taken into account.

(d) for uniform or non uniform flow since u, v, w are local variables at any point.

(a) Steady Compressible flow

Steady $\Rightarrow \frac{\partial \rho}{\partial t} = 0$

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

(b) Steady incompressible flow

Incompressible $\Rightarrow \rho = \text{Constant}$

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial y} = \frac{\partial \rho}{\partial z} = \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

(c) Two-dimensional flow

$w = 0$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Existence of flow

If a fluid flow with velocity field $\vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$ were to exist, the velocity components must conform with mass conservation principle i.e. for possible flow the velocity components must obey continuity equation.

Tutorial Problem (Solved)

Two components of velocity in an incompressible ^{fluid} flow are given as

$$u = x^2 - y^2 \quad \text{and} \quad v = z^2 - y^2$$

Determine the third component, assuming that the origin is a stagnation point.

Solution

The Continuity eqⁿ requires that

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$(2x^2 - 0) + (0 - 2y^2) + \frac{\partial w}{\partial z} = 0$$

$$\frac{\partial w}{\partial z} = 2y^2 - 2x^2 = 2(y^2 - x^2)$$

$$w = 2(y^2 - x^2)z + f(x, y)$$

↳ Constant of integration, independent

of z but may be function of x and y .

origin is a stagnation point

i.e. at $(0, 0, 0)$, $u = v = w = 0$

$$w = 0, \quad \text{at } (0, 0, 0)$$

$$\Rightarrow f(x, y) = 0$$

$$\boxed{w = 2(y^2 - x^2)z}$$

LECTURE 17

Rotation and Vorticity

"

The Rotation of a fluid particle at a point is given by average angular velocity of two differential linear elements of fluid particle originally perpendicular to each other."

— Consider rectangular element & Rect of fluid. Dotted line shows position of rectangular & Rect after time δt .

— At time t point A has velocities u and v in x and y directions respectively.

— Point D has velocity

$(u + \frac{\partial u}{\partial y} \delta y)$ in x -direction

— Point B has velocity $u + \frac{\partial u}{\partial x} \delta x$ in x -direction.

— Due to difference in velocity of A and D points, linear element AD will have angular velocity.

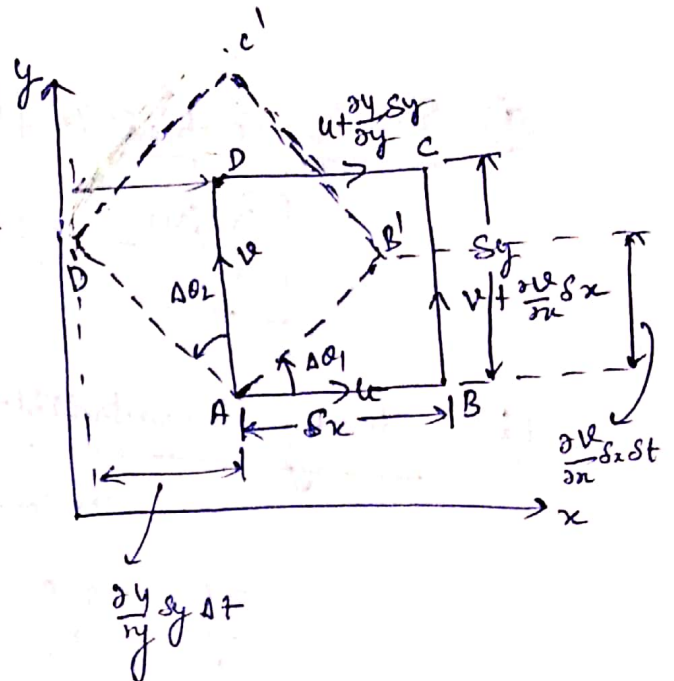
— Similarly due to difference in velocity of A and B points, linear element AB will have angular velocity.

$$\text{Angular velocity of AB} = \frac{\Delta \theta_1}{\delta t} = \frac{\frac{\partial v}{\partial x} \delta x \delta t}{\delta x \delta t} = \frac{\partial v}{\partial x}$$

$$\text{Angular velocity of AD} = \frac{\Delta \theta_2}{\delta t} = \frac{-\frac{\partial u}{\partial y} \delta y \delta t}{\delta y \delta t} = -\frac{\partial u}{\partial y}$$

Rotation about z -axis (motion being in x - y plane)

$$\boxed{\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)}$$



If the flow is three dimensional, then

Similarly we can write

Rotation about x-axis $w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$

Rotation about y-axis $w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$

These three scalar equations are put together and written in vector form as

$$\vec{\omega} = i w_x + j w_y + k w_z$$

$$= \frac{1}{2} (\nabla \times \vec{V})$$

$$= \frac{1}{2} \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$= \frac{1}{2} \text{curl } \vec{V}$$

$$(\vec{V} = u + vj + wk)$$

$$(\nabla \times \vec{V} = \text{curl of velocity vector})$$

Vorticity: Vorticity in a fluid motion is taken numerically equal to twice the value of rotation.

$$\Omega = 2 \vec{\omega} \Rightarrow \vec{\omega} = \frac{\Omega}{2}$$

$$\text{Rotation} = \frac{1}{2} \text{vorticity}$$

For irrotational flow

each component of rotation or vorticity must vanish

i.e. $w_x = w_y = w_z = 0$

For 2-D flow in xy plane

$$w_z = 0 \Rightarrow \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = 0 \Rightarrow \boxed{\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}}$$

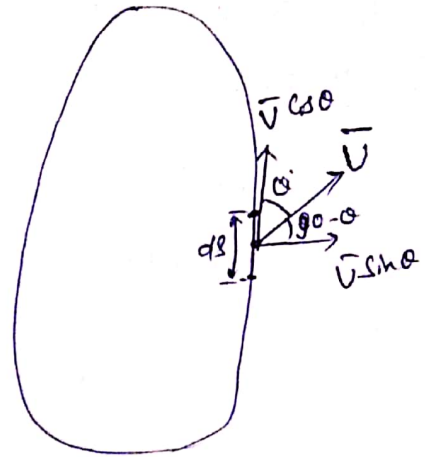
Circulation and vorticity

"Circulation (Γ) is defined as line integral of the tangential velocity of fluid around a closed contour in the flow field."

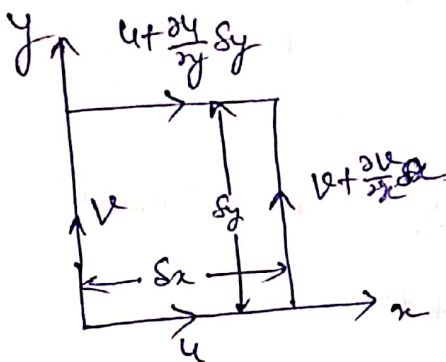
$$\Gamma = \oint \vec{V} \cdot d\vec{s}$$

- $\oint \Rightarrow$ Summation around a closed contour.

- By convention circulation is considered positive in anticlockwise direction.



Consider rectangular fluid element with dimensions s_x and s_y



$$\vec{U} = u\vec{i} + v\vec{j} + w\vec{k}$$

$$d\vec{s} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\Gamma = \oint \vec{U} \cdot d\vec{s}$$

for three-D flow $\Gamma = \oint (u dx + v dy + w dz)$

For two-D flow $\Gamma = \oint u dx + v dy$

$$\delta \Gamma = u s_x + (v + \frac{\partial v}{\partial x} s_x) s_y - (u + \frac{\partial u}{\partial y} s_y) s_x - v s_y$$

$$= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) s_x s_y$$

$$\delta \Gamma = 2 \omega_z \delta A$$

$$\frac{\delta \Gamma}{\delta A} = 2 \omega_z = \Omega = \text{Vorticity}$$

$\delta A =$ Contour area in a plane right angle to z -axis.

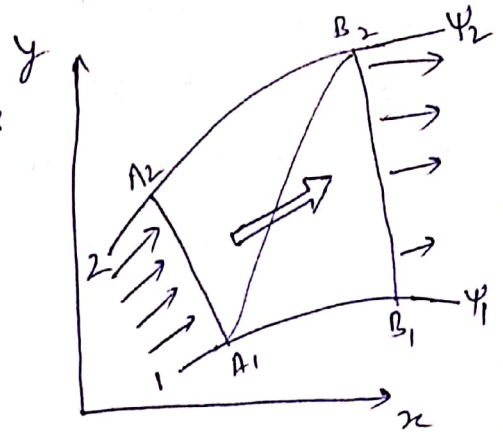
- Hence, circulation per unit area equals vorticity in the flow.

LECTURE 18

Stream function (Ψ)

- Since no fluid can cross a streamline, the flow occurring between two streamlines must remain unchanged.

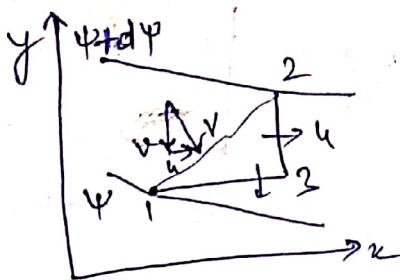
$$\begin{aligned} \text{i.e. flow across } A_1 A_2 &= \text{flow across } B_1 B_2 \\ &= \text{flow across } A_1 B_2 \end{aligned}$$



- A value of stream function Ψ is assigned to each streamline such that volumetric flow rate is given by $\Psi_2 - \Psi_1$.
- Hence streamlines are lines of constant stream function.
- Unit of Ψ is m^2/s (discharge per unit thickness of flow).
- mathematically: Stream function for a flow in x - y plane is defined as a ^{scalar} function of x and y such that

$$u = \frac{\partial \Psi}{\partial y}, \quad v = -\frac{\partial \Psi}{\partial x}$$

PHYSICAL CONCEPT



- Two neighbouring streamlines defined by stream function Ψ and $\Psi + d\Psi$

$$\Psi = \Psi(x, y)$$

Total differential

$$d\Psi = \frac{\partial \Psi}{\partial x} dx + \frac{\partial \Psi}{\partial y} dy$$

Using mathematical definition of Ψ

$$\begin{aligned} \text{flow across 1-2} &= d\Psi = \underbrace{-v dx}_{\text{flow across 1-3}} + \underbrace{u dy}_{\text{flow across 2-3}} \end{aligned}$$

(-ve sign indicates velocity is opposite to direction)

Checking continuity eqn in two dimension for Cartesian coordinates

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad \frac{\partial}{\partial x} \left(\frac{\partial \Psi}{\partial y} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial \Psi}{\partial x} \right) = \frac{\partial^2 \Psi}{\partial x \partial y} - \frac{\partial^2 \Psi}{\partial x \partial y} = 0 \quad \text{Satisfied}$$

Hence for a possible flow stream function (Ψ) must exist.

Velocity potential

"A function $\phi = \phi(x, y)$ is defined such that its partial derivative with respect to ~~any~~ displacement in any direction is equal to velocity component in that direction."

mathematically

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}, \quad w = \frac{\partial \phi}{\partial z}$$

$$\text{or } \boxed{\nabla \phi = \vec{V}}$$

for 2-D flow

$$\frac{\partial \phi}{\partial x} = u, \quad \frac{\partial \phi}{\partial y} = v$$

Checking condition of irrotationality

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

$$\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) = \frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} = 0$$

- Hence if a ^{possible} flow is irrotational, velocity potential ϕ must exist.

~~Two~~ "Two dimensional irrotational flows are also called potential flow."

Laplace equations

- Substituting ψ in irrotationality condition

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \Rightarrow \frac{\partial}{\partial x} \left(-\frac{\partial \psi}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial y} \right) \Rightarrow \boxed{\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0}$$

Hence stream function for irrotational flow satisfies Laplace equation.

- Substituting ϕ with continuity equation.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) = \boxed{\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0}$$

Hence velocity potential satisfies Laplace equation.

TUTORIAL 5

Tutorial problem (solved)

For the following flows, determine the components of rotation about the various axes.

(i) $u = xy^3z$, $v = -y^2z^2$, $w = yz^2 - \frac{y^3z^2}{2}$

(ii) $u = 3xy$, $v = \frac{3}{2}x^2 - \frac{3}{2}y^2$

(iii) $u = y^2$, $v = -3x$

Solution

Components of rotation about various axes

Rotation about Z-axis $\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$

- " - x-axis $\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$

- " - y-axis $\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$

(i) $\omega_z = \frac{1}{2} (0 - 3xy^2z) = -\frac{3}{2}xy^2z$

$$\omega_x = \frac{1}{2} (z^2 - \frac{3}{2}y^2z^2 + 2y^2z)$$

$$\omega_y = \frac{1}{2} (xy^3 - 0) = \frac{1}{2}xy^3$$

(ii) $\omega_z = \frac{1}{2} (3x - 3x) = 0$

∴ as the flow is having only u and v components.
i.e. flow is two dimensional in xy plane

$$\underline{\underline{\omega_x = \omega_y = 0}}$$

(iii) $\omega_z = \frac{1}{2} (-3 - 2y)$

flow is two dimensional in xy plane

$$\underline{\underline{\omega_x = \omega_y = 0}}$$

Tutorial problem (Solved)

(i) The velocity distribution for a three dimensional flow is

$$u = a + by - cz$$

$$v = d - bx - ez$$

$$w = f + cx - ey$$

where a, b, c, d, e and f are arbitrary constants. Show that these are the velocity components of a fluid motion?

(ii) Does these velocity components represent irrotational flow? If not, determine vorticity and rotation.

Solution

(i) For a possible fluid motion, continuity eqn should be satisfied by velocity components

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$0 + 0 + 0 = 0 \quad (\text{satisfied})$$

Hence flow is possible

(ii) For irrotational flow each component of rotation must be zero,

$$\omega_x = \omega_y = \omega_z = 0$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (-b - b) = -b$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) = \frac{1}{2} (-e - (-e)) = 0$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) = \frac{1}{2} (-c - c) = -c$$

$$\vec{\omega} = i\omega_x + j\omega_y + k\omega_z = \underline{\underline{-bj - ck}}$$

$$\omega = \sqrt{b^2 + c^2}$$

$$\text{Vorticity} = 2 \cdot \text{rotation}$$

$$\text{Vorticity} = \underline{\underline{2\sqrt{b^2 + c^2}}}$$

Tutorial Problem (Solved)

The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as

$$u = y^2 - x^2 + y$$

$$v = 2xy + x$$

- (i) Show that these functions represent a possible case of an irrotational flow
- (ii) obtain an expression for Stream function ψ .
- (iii) obtain an expression for velocity potential ϕ .

Solution

- (i) For possible flow $\boxed{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0}$
2D Continuity eqⁿ should be satisfied

$$-2x + 2x = 0$$

(Hence flow is possible)

for irrotational flow

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

$$(2y + 1) - (2y + 1) = 0$$

(Hence flow is irrotational)

$$v = -\frac{\partial \psi}{\partial x}$$

$$-(2xy + x) = +\frac{\partial \psi}{\partial x}$$

$$-(x^2y + \frac{x^2}{2}) + g(y) = \psi$$

$$y^2 - x^2 + y = \frac{\partial \psi}{\partial y}$$

$$\frac{y^3}{3} - x^2y + \frac{y^2}{2} + f(x) = \psi$$

inspecting eqⁿ.

$$\boxed{\psi = \frac{y^3}{3} - x^2y + \frac{y^2}{2} + \frac{x^2}{2}}$$

$$\textcircled{\text{III}} \quad u = \frac{\partial \phi}{\partial x} \quad \left| \quad v = \frac{\partial \phi}{\partial y} \right.$$

$$y^2 - x^2 + y = \frac{\partial \phi}{\partial x} \quad \left| \quad 2xy + x = \frac{\partial \phi}{\partial y} \right.$$

$$xy^2 - \frac{x^3}{3} + xy + f(y) = \phi \quad \left| \quad \underline{xy^2 + xy + f(x)} = \phi \right.$$

$$\phi = \underline{\underline{xy^2 + xy - \frac{x^3}{3}}}$$

Alternatively

Note! $u = \frac{\partial \psi}{\partial y} \Rightarrow y^2 - x^2 + y = \frac{\partial \psi}{\partial y}$

$$\Rightarrow \psi = \frac{y^3}{3} - x^2 y + \frac{y^2}{2} + f(x)$$

$$v = -\frac{\partial \psi}{\partial x} \Rightarrow \frac{\partial \psi}{\partial x} = -2xy + f'(x)$$

$$2xy + x = -\frac{\partial \psi}{\partial x} \longrightarrow = -2xy - x = -xy + f'(x)$$

$$f'(x) = -x$$

$$f(x) = -\frac{x^2}{2}$$

$$\boxed{\psi = \frac{y^3}{3} - x^2 y + \frac{y^2}{2} - \frac{x^2}{2}}$$

$$u = \frac{\partial \phi}{\partial x} \Rightarrow y^2 - x^2 + y = \frac{\partial \phi}{\partial x}$$

$$\Rightarrow \phi = y^2 x - \frac{x^3}{3} + xy + f(y)$$

$$v = \frac{\partial \phi}{\partial y} = 2yx + x + f'(y)$$

$$2xy + x = 2yx + x + f'(y)$$

$$f'(y) = 0$$

$$f(y) = 0$$

$$\boxed{\phi = y^2 x - \frac{x^3}{3} + xy}$$

Tutorial problem (Solved)

Verify whether the following functions are valid potential functions.

(i) $\phi = Axy$ (ii) $\phi = m \ln x$ (iii) $\phi = A(x^2 - y^2)$ (iv) $\phi = A \cos x$

Solution

for a valid potential function it must satisfy Laplace equation.

$$(i) \quad \phi = Axy, \quad \left. \begin{array}{l} \frac{\partial \phi}{\partial x} = Ay, \quad \frac{\partial^2 \phi}{\partial x^2} = 0 \\ \frac{\partial \phi}{\partial y} = Ax, \quad \frac{\partial^2 \phi}{\partial y^2} = 0 \end{array} \right\} \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (\text{Satisfied})$$

Hence $\phi = Axy$ is valid potential function.

$$(ii) \quad \phi = m \ln x, \quad \frac{\partial \phi}{\partial x} = \frac{m}{x}, \quad \frac{\partial^2 \phi}{\partial x^2} = -\frac{m}{x^2}$$
$$\frac{\partial \phi}{\partial y} = 0, \quad \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$\Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\frac{m}{x^2} \neq 0 \quad (\text{not Satisfied})$$

Hence, $\phi = m \ln x$ is not a valid potential function.

$$(iii) \quad \phi = A(x^2 - y^2), \quad \frac{\partial \phi}{\partial x} = 2Ax, \quad \frac{\partial \phi}{\partial y} = -2Ay$$
$$\frac{\partial^2 \phi}{\partial x^2} = 2A, \quad \frac{\partial^2 \phi}{\partial y^2} = -2A$$

$$\Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (\text{Satisfied})$$

Hence, $\phi = A(x^2 - y^2)$ is a valid potential function.

$$(iv) \quad \phi = A \cos x; \quad \frac{\partial \phi}{\partial x} = -A \sin x, \quad \frac{\partial^2 \phi}{\partial x^2} = -A \cos x$$
$$\frac{\partial \phi}{\partial y} = 0, \quad \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$\Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \neq 0 \quad (\text{not Satisfied})$$

Hence, $\phi = A \cos x$ is not a valid potential function.

Tutorial problem (solved) Check whether the flow defined by stream function $\psi = 2xy$ is irrotational? If so determine the corresponding velocity potential.

Solution:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

for flow to be irrotational stream function must satisfy Laplace eqⁿ.

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} = 0$$

$$\frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\frac{\partial \psi}{\partial y} = 2x$$

$$\frac{\partial \psi}{\partial x} = 2y$$

$\Rightarrow$ Laplace eqⁿ is satisfied
flow is irrotational.

$$u = 2x, \quad v = -2y$$

$$\frac{\partial \phi}{\partial x} = u, \quad \frac{\partial \phi}{\partial y} = v$$

$$\frac{\partial \phi}{\partial x} = u \Rightarrow \frac{\partial \phi}{\partial x} = 2x \Rightarrow$$

$$\phi = x^2 + f(y)$$

$$\frac{\partial \phi}{\partial y} = 0 + f'(y)$$

$$-2y = 0 + f'(y)$$

$$f'(y) = -2y$$

$$f(y) = -y^2$$

$$\phi = x^2 + f(y)$$

$$\boxed{\phi = x^2 - y^2} \text{ Ans}$$

Tutorial problem (solved) The flow field is characterised by $\psi = x^3y$. Determine the velocity potential ϕ for the flow.

Solution

for the flow to be irrotational

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\frac{\partial \psi}{\partial x} = 3x^2y$$

$$\frac{\partial^2 \psi}{\partial x^2} = 6xy$$

$$\frac{\partial \psi}{\partial y} = x^3$$

$$\frac{\partial^2 \psi}{\partial y^2} = 0$$

Hence Laplace equation is not satisfied by stream function.

Hence flow is not irrotational

Hence velocity potential cannot be defined for this flow.